Marginal Product: Practice Questions

Economics — Production Theory

Instructions: Answer all questions. Show all workings clearly.

Question 1: Marginal Product from a Table (Midpoint Method)

Thandi is preparing for her final economics examination. The table below shows the relationship between her total study hours (the input) and her expected final mark out of 100 (the output). Note that study hours are recorded in unequal intervals.

Study Hours (H)	Final Mark (Q)	Marginal Product (MP)
0	12	
8	28	
16	48	
24	64	
32	72	
40	82	
48	88	
56	90	

Answer the following:

(a) Calculate the marginal product of 32nd hour of study. Complete the third column of the table above. Round your answers to two decimal places.

Question 2: Deriving Marginal Product from a Total Product Function

Sipho is studying for his statistics final. His total product function — which gives his expected final mark (Q) as a function of hours studied (H) — is estimated as:

$$Q(h) = -0.3h^2 + 9h + 15$$

where Q is the final mark (out of 100) and h is the number of hours studied.

Answer the following:

(a) Derive the marginal product function MP(h).

(b) Calculate the marginal product of the 10th hour of study. Interpret your answer in the context of Sipho's exam preparation.

Memorandum (Answer Key)

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Question 1 — Answers

(a) Marginal Product calculations (midpoint method):

Interval	ΔH	ΔQ	MP = ΔQ/ΔH	Midpoint H
0 → 3	3	16	5.33	1.5
3 → 7	4	20	5.00	5.0
7 → 12	5	16	3.20	9.5
12 → 15	3	8	2.67	13.5
15 → 20	5	10	2.00	17.5
20 → 24	4	6	1.50	22.0
24 → 30	6	2	0.33	27.0

(b) The marginal product is highest between 0 and 3 hours (MP = 5.33 marks per hour). This means the first few hours of study are the most productive — each additional hour yields the greatest improvement in Thandi's mark.

(c) The marginal product begins to decline after the 0→3 interval. From the second interval onward (3→7, MP = 5.00), each additional hour produces fewer additional marks. This illustrates the law of diminishing marginal returns: holding all other inputs constant (e.g., textbook, ability, rest), each successive hour of study adds less to the total mark.

(d) MP from 20 to 24 hours = (88 − 82) / (24 − 20) = 6/4 = 1.50 marks per hour. Over 4 hours, Thandi gains 6 marks. Cost of 4 extra hours = 4 × R50 = R200. Benefit of 6 extra marks = 6 × R120 = R720. Since the benefit (R720) exceeds the cost (R200), the extra study is worthwhile.

(e) Graph should show a downward-sloping MP curve (marks per hour on y-axis, study hours on x-axis). The entire curve beyond the first interval should be labelled as the region of diminishing marginal returns. MP remains positive throughout but declines continuously.

Question 2 — Answers

(a) MP(H) = dQ/dH = −0.6H + 9

(b) MP(10) = −0.6(10) + 9 = −6 + 9 = 3.0 marks per hour. At the 10th hour of study, Sipho gains an additional 3 marks. While studying is still productive, the return is diminishing compared to earlier hours.

(c) MP(16) = −0.6(16) + 9 = −9.6 + 9 = -0.6 marks per hour. The marginal product is negative, meaning that studying beyond this point actually reduces Sipho's mark — likely due to exhaustion, burnout, or confusion from over-studying.

(d) Set $MP(H) = 0$: $-0.6H + 9 = 0 \rightarrow H = 9/0.6 = 15$ hours. At 15 hours, the marginal product equals zero. This is the point where total product is maximised — any additional studying beyond this point would be counterproductive.

(e) $Q(15) = -0.3(15^2) + 9(15) + 15 = -0.3(225) + 135 + 15 = -67.5 + 135 + 15 = 82.5$. Yes, this is the maximum possible mark. Since $MP = 0$ at $H = 15$, the total product function reaches its peak here. Beyond 15 hours, $MP < 0$ and the mark would decrease.

(f) MP is positive but diminishing for all hours from $H = 0$ to $H = 15$. Since the MP function is linear and downward-sloping ($MP = -0.6H + 9$), it is always diminishing. The law of diminishing marginal returns applies from the very first hour — each successive hour adds less to the total mark than the previous one, until the 15th hour where $MP = 0$.

(g) $MP(9) = -0.6(9) + 9 = 3.6$ marks. $MP(10) = -0.6(10) + 9 = 3.0$ marks. Total additional marks from hours 9 and 10 = $3.6 + 3.0 = 6.6$ marks. The social event is worth 5 marks equivalent.

- Hour 9 alone yields 3.6 marks > 5 (opportunity cost), so studying hour 9 is worthwhile.
- Hour 10 yields 3.0 marks < 5 (opportunity cost), so studying hour 10 is NOT worthwhile.

Sipho should study for one more hour (hour 9) and then go to the social event. This is a direct application of the marginal decision rule: continue an activity as long as the marginal benefit exceeds the marginal cost.